Introduction to Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation (RAG) is an AI architecture that combines information retrieval with large language models (LLMs). Instead of relying only on the model's internal knowledge, RAG retrieves relevant information from external documents before generating a response.

Key Components

1. Document Loader: Reads files such as PDFs.
2. Text Splitter: Divides documents into smaller chunks.
3. Embedding Model: Converts text into vector embeddings.
4. Vector Database: Stores embeddings for efficient similarity search.
5. Retriever: Finds the most relevant chunks based on a user query.
6. Large Language Model: Uses the retrieved context to generate accurate answers.

Advantages of RAG

- Reduces hallucinations.
- Uses up-to-date external knowledge.
- Improves factual accuracy.
- Works well for question-answering over custom documents.

Example Questions

• What is Retrieval-Augmented Generation?
• What is the purpose of embeddings?
• Why is FAISS used in a RAG system?
• Name the main components of a RAG pipeline.